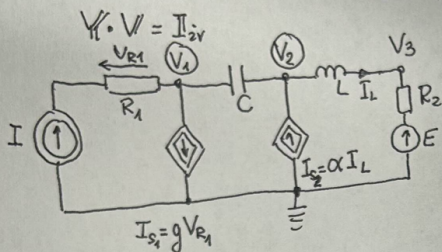


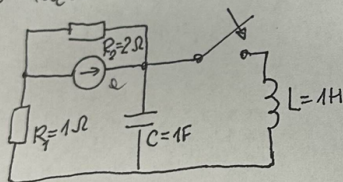
Examin TOIS 17.06.25

(A)

1. Opisać obwód równaniem napięć w formie macierzy

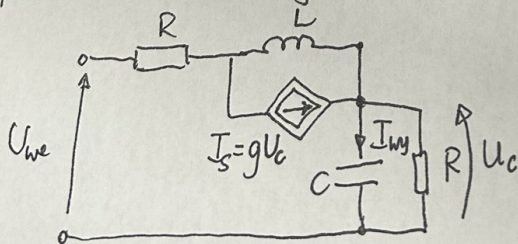


2. Obkreślić $u_C(t)$ oraz $i_L(t)$ w stanie niestabilnym po zamknięciu wyłącznika



$e(t) = 20V$

3. Obkreślić transmitancję $H(s) = \frac{I_{wy}(s)}{U_{we}(s)}$ obrotu, odpowiedź impulsową oraz chętyli cyfrowe.



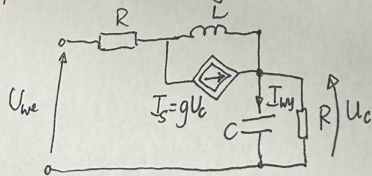
$R = 1\Omega$

$L = 1H$

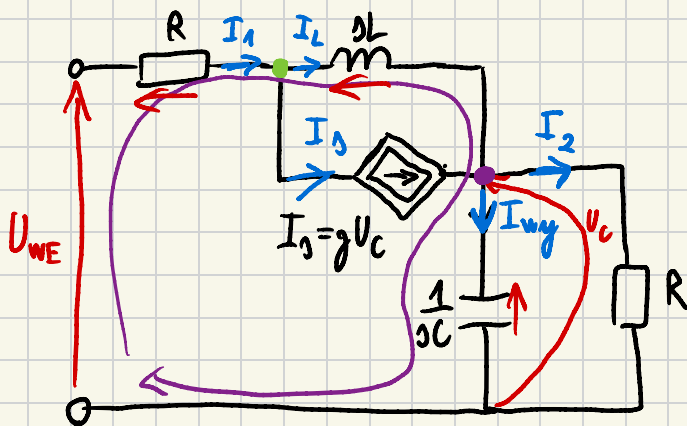
$C = 1F$

$g = 3$

3. Obkresliť transmittanciu $H(s) = \frac{I_{wy}(s)}{U_{we}(s)}$ obvodu, odpovedí impulzom oraz chytily cyfroliswówwe.



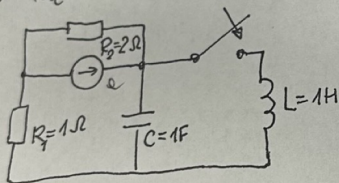
$$\begin{aligned} R &= 1 \Omega \\ L &= 1 \text{ H} \\ C &= 1 \text{ F} \\ g &= 3 \end{aligned}$$



$$\begin{aligned} U_c &= I_2 \cdot R \quad ! \\ I_2 &= \frac{U_c}{R} \end{aligned}$$

$$I_{S_1} = g V_{R_1}$$

2. Określić $u_C(t)$ oraz $i_L(t)$ w stanie niestabilnym po zamknięciu wyłącznika

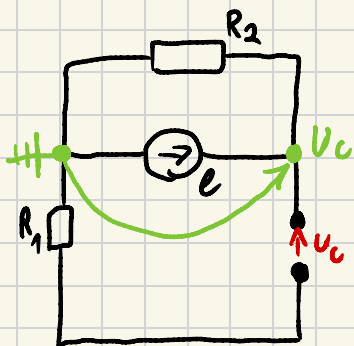


$$e(t) = 20V$$

$$R_1 = 1 \quad R_2 = 2$$

$$C = 1 \quad L = 1$$

1° Warunki początkowe

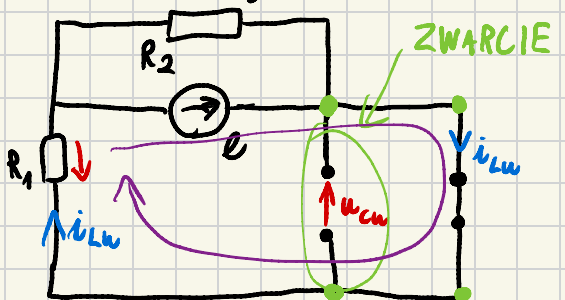


$$i_L(t) = 0 \text{ A} \quad i_L(0^-) = 0 \text{ A}$$

$$u_C(t) = e = 20 \text{ [V]}$$

$$u_C(0^-) = 20 \text{ [V]}$$

2° Stan ustalony po przełączeniu



$$u_{Cw}(t) = 0 \text{ [V]}$$

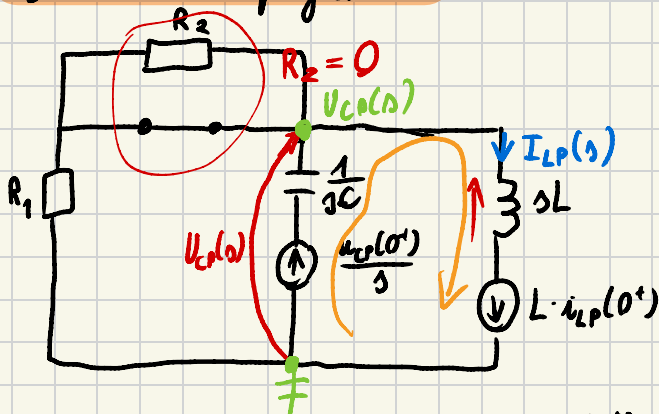
$$u_{Cw}(0^+) = 0 \text{ [V]}$$

$$e(t) - i_{Lw} \cdot R_1 = 0$$

$$i_{Lw}(t) = \frac{e(t)}{R_1} = \frac{20}{1} = 20 \text{ [A]}$$

$$i_{Lw}(0^+) = 20 \text{ [A]}$$

3° Skladowna pręciwna



$$u_c(0^-) = u_{cu}(0^+) + u_{cp}(0^+)$$

$$u_{cp}(0^+) = u_c(0^-) - u_{cu}(0^+)$$

$$u_{cp}(0^+) = 20 - 0 = 20 [V]$$

$$i_{LP}(0^+) = i_L(0^-) - i_{Lu}(0^+)$$

$$i_{LP}(0^+) = 0 - 20 = -20 [A]$$

$$U_{CP}(s) \left[\frac{1}{R_1} + sL + \frac{1}{sL} \right] = \frac{u_{CP}(0^+)}{s} - L \cdot i_{LP}(0^+)$$

$$U_{CP}(s) \left[1 + s + \frac{1}{s} \right] = \frac{20}{s} + 20 \parallel s$$

$$U_{CP}(s) [s^2 + s + 1] = 20s + 20$$

$$U_{CP}(s) = 20 \frac{s+1}{s^2+s+1} \rightarrow \Delta < 0$$

$$U_{CP}(s) = 20 \frac{s + \frac{1}{2} + \frac{1}{4}}{s^2 + s + \frac{1}{4} + \frac{3}{4}} = 20 \frac{s + \frac{1}{2} + \frac{1}{4}}{(s + \frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} =$$

$$= 20 \left[\frac{s + \frac{1}{2}}{(s + \frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} + \frac{1}{2} \frac{1}{(s + \frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} \right] = 20 \left[\frac{s + \frac{1}{2}}{(s + \frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} + \frac{1}{2} \cdot \frac{\frac{\sqrt{3}}{2} \cdot \frac{2}{\sqrt{3}}}{(s + \frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} \right] =$$

$$u_{CP}(t) = 20 \left[e^{-\frac{1}{2}t} \cos\left(\frac{\sqrt{3}}{2}t\right) + \frac{1}{2} \cdot \frac{2}{\sqrt{3}} \cdot e^{-\frac{1}{2}t} \sin\left(\frac{\sqrt{3}}{2}t\right) \right] =$$

$$= 20 e^{-\frac{1}{2}t} \cos\left(\frac{\sqrt{3}}{2}t\right) + \frac{20\sqrt{3}}{3} e^{-\frac{1}{2}t} \sin\left(\frac{\sqrt{3}}{2}t\right)$$

$$u_c(t) = u_{cu}(t) + u_{cp}(t) = 20 e^{-\frac{1}{2}t} \cos\left(\frac{\sqrt{3}}{2}t\right) + \frac{20\sqrt{3}}{3} e^{-\frac{1}{2}t} \sin\left(\frac{\sqrt{3}}{2}t\right)$$

TABLICA
TRANSFORMAT

IM NPK

$$U_{CP}(s) - I_{LP}(s) \cdot sL + L \cdot i_{LP}(0^+) = 0$$

$$20 \cdot \frac{s+1}{s^2+s+1} - I_{LP}(s) \cdot s - 20 = 0$$

$$I_{LP}(s) \cdot s = 20 \left(\frac{s+1}{s^2+s+1} - 1 \right)$$

$$I_{LP}(s) \cdot s = 20 \left(\frac{\cancel{s+1} - s^2 - s - 1}{s^2+s+1} \right)$$

$$I_{LP}(s) \cdot s = 20 \left(\frac{-s^2}{s^2+s+1} \right)$$

$$I_{LP}(s) = -20 \frac{s}{s^2+s+1} = -20 \frac{s+\frac{1}{2}-\frac{1}{2}}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2} =$$

$$= -20 \left[\frac{s+\frac{1}{2}}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2} - \frac{1}{2} \frac{1}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2} \right] =$$

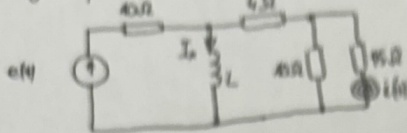
$$= -20 \left[\frac{s+\frac{1}{2}}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2} - \frac{1}{2} \frac{\frac{\sqrt{3}}{2} \cdot \frac{2}{\sqrt{3}}}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2} \right]$$

$$i_{LP}(t) = -20 \left[e^{-\frac{1}{2}t} \cos(\frac{\sqrt{3}}{2}t) - \frac{1}{2} \cdot \frac{2}{\sqrt{3}} \cdot e^{-\frac{1}{2}t} \sin(\frac{\sqrt{3}}{2}t) \right] =$$

$$= -20 e^{-\frac{1}{2}t} \cos(\frac{\sqrt{3}}{2}t) + \frac{20\sqrt{3}}{3} e^{-\frac{1}{2}t} \sin(\frac{\sqrt{3}}{2}t)$$

$$i_L(t) = i_{LW}(t) + i_{LP}(t) = 20 - 20 e^{-\frac{1}{2}t} \cos(\frac{\sqrt{3}}{2}t) + \frac{20\sqrt{3}}{3} e^{-\frac{1}{2}t} \sin(\frac{\sqrt{3}}{2}t)$$

1. Obtiayt $i_L(t)$ metodu Thevenina

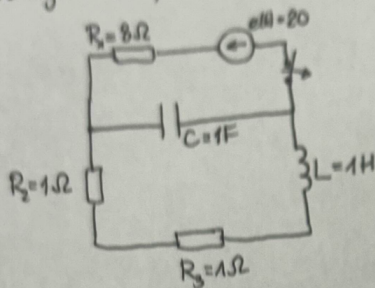


$$i(t) = 2.88 \sin(\omega t - 90^\circ)$$

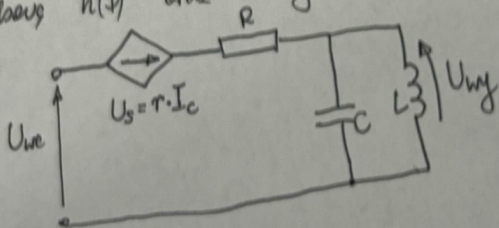
$$e(t) = 40.28 \sin(\omega t + 45^\circ)$$

$$\omega L = 5 \Omega$$

2. Obtiayt $u_C(t)$, $i_L(t)$ na stavu nistatolompa po otvornu vytkivaniya



3. Okreidite transformatsiyu $H(s)$ obredu, $H(s) = \frac{U_{vy}(s)}{U_{ve}(s)}$, odpovedi impulsu $\delta(t)$ oraz ch-tyli ustakivirovane.



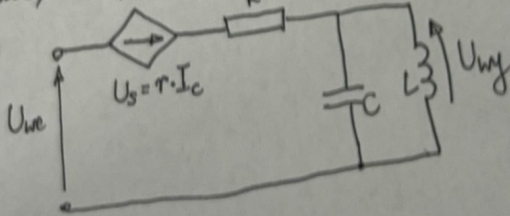
$$R = 2 \Omega$$

$$L = 1H$$

$$C = 1F$$

$$r = 3$$

3. Określić transmitancję $H(s)$ obwodu, $H(s) = \frac{U_{wy}(s)}{U_{we}(s)}$, odpowiedź impulsową $h(t)$ oraz częstotli charakterystykę.

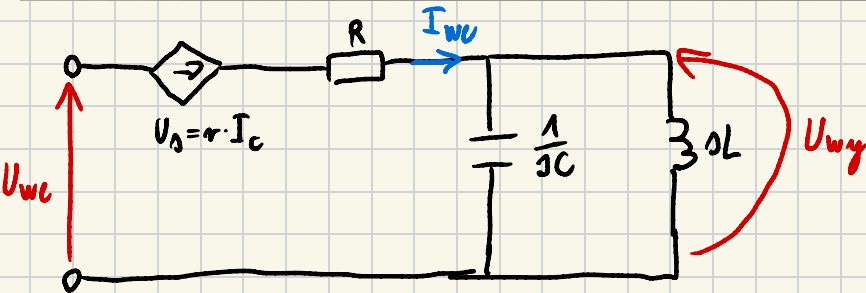


$$R = 2 \Omega$$

$$L = 1 \text{ mH}$$

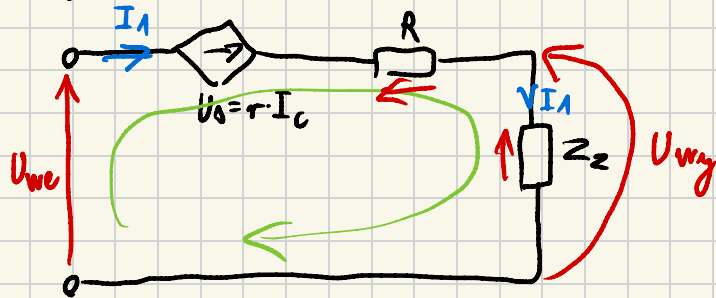
$$C = 1 \text{ F}$$

$$r = 3$$



$$I_c = \frac{U_{wy}}{\frac{1}{sC}}$$

$$I_c = sC \cdot U_{wy}$$



$$Y_z = sC + \frac{1}{sL}$$

$$Y_z = s + \frac{1}{s} = \frac{s^2 + 1}{s}$$

$$Z_z = \frac{1}{Y_z} = \frac{s}{s^2 + 1}$$

$$U_{we} - I_1 \cdot R - I_1 \cdot Z_z + r I_c = 0$$

$$I_1 = \frac{U_{wy}}{Z_z}$$

$$U_{we} - \frac{U_{wy}}{Z_z} \cdot R - \frac{U_{wy}}{Z_z} \cdot Z_z + r \cdot sC \cdot U_{wy} = 0$$

$$\frac{U_{wy}}{U_{we}} = \frac{s}{-s^2 + s + 2}$$

$$U_{we} - \frac{U_{wy}}{Z_z} \cdot 2 - U_{wy} + 3s \cdot U_{wy} = 0$$

$$U_{we} = U_{wy} \left[\frac{2}{Z_z} + 1 - 3s \right]$$

$$U_{we} = U_{wy} \left[2 \cdot \frac{s^2 + 1}{s} + 1 - 3s \right]$$

$$U_{we} \cdot s = U_{wy} [2s^2 + 2 + s - 3s^2]$$

$$H(s) = \frac{s}{-s^2 + s + 2}$$

Odp. impulsowa

$$H(s) = \frac{s}{-s^2 + s + 2} = -\frac{s}{s^2 - s - 2} = \frac{-s}{(s+1)(s-2)}$$

$$s^2 - s - 2 = 0$$

$$\Delta = 9 \quad \sqrt{\Delta} = 3$$

$$s_1 = \frac{1-3}{2} = -1 \quad s_2 = 2$$

$$y(t) = \lim_{s \rightarrow -1} \left(\frac{-s}{s-2} \cdot e^{st} \right) + \lim_{s \rightarrow 2} \left(\frac{-s}{s+1} \cdot e^{st} \right) =$$

$$= \frac{1}{-3} \cdot e^{-t} + \frac{-2}{3} \cdot e^{2t} = -\frac{1}{3} e^{-t} - \frac{2}{3} e^{2t}$$

Charakterystyki częstotliwościowe

$$H(j\omega) = \frac{-j\omega}{(j\omega)^2 - j\omega - 2} = \frac{0 - j\omega}{-2 - \omega^2 - j\omega} = \frac{j\omega}{2 + \omega^2 + j\omega}$$

$$A(\omega) = 0 \quad B(\omega) = \omega$$

$$C(\omega) = 2 + \omega^2 \quad D(\omega) = \omega$$

$$\begin{aligned} H(s=j\omega) &= \frac{A(\omega) + jB(\omega)}{C(\omega) + jD(\omega)} = H(\omega) \\ |H(\omega)| &= \frac{|A(\omega) + jB(\omega)|}{|C(\omega) + jD(\omega)|} \rightarrow \text{amplitudowa charakterystyka} \\ \varphi(\omega) &= \arctg\left(\frac{B(\omega)}{A(\omega)}\right) - \arctg\left(\frac{D(\omega)}{C(\omega)}\right) \rightarrow \text{fazowa charakterystyka} \end{aligned}$$

$$|H(\omega)| = \frac{\sqrt{0^2 + \omega^2}}{\sqrt{(2 + \omega^2)^2 + \omega^2}} = \frac{\omega}{\sqrt{\omega^2 + (2 + \omega^2)^2}} \leftarrow \text{amplitudowa}$$

$$\varphi(\omega) = \arctg\left(\frac{\omega}{0}\right) - \arctg\left(\frac{\omega}{2 + \omega^2}\right) \leftarrow \text{fazowa}$$